

C++ Decision Making Practice Questions

For each question, trace the execution of the code and determine the output printed to the console, including the final values of the variables a, b, c, and d. Pay close attention to short-circuit evaluation (&&, ||) and pre/post increment/decrement operators (++ , --)

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Programming Exercises

Question 1

```
int a = 0, b = 1, c = 2, d = 3;
```

```
if (a++ && b++) cout << "Apple";
```

```
else if (a++ || c++) cout << "Banana";
```

```
else cout << "Cherry";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 2

```
int a = 1, b = 1, c = 1, d = 1;
```

```
if (--a || --b || --c) cout << "Red";
```

```
else if (d--) cout << "Blue";
```

```
else cout << "Green";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 3

```
int a = 2, b = 0, c = -2, d = 0;

if (b++ && c++) cout << "One";

else if (++b && ++a) cout << "Two";

else cout << "Three";

cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 4

```
int a = -1, b = -1, c = -1, d = -1;

if (++a && ++b) cout << "A";

else if (a++ || b++) cout << "B";

else cout << "C";

cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 5

```
int a = 0, b = 0, c = 0, d = 0;

if (a++ || b++ || c++ || d++) cout << "Yes";

else cout << "No";

cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 6

```
int a = 5, b = 5, c = 5, d = 5;

if (a-- && b-- && c-- && d--) cout << "All";
```

```
else cout << "None";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 7

```
int a = 1, b = 2, c = 3, d = 4;
```

```
if (--a && ++b) cout << "X";
```

```
else if (c-- && --d) cout << "Y";
```

```
else cout << "Z";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 8

```
int a = 0, b = 1, c = 0, d = 1;
```

```
if (a++ && b++) cout << "P";
```

```
else if (c++ || --d) cout << "Q";
```

```
else if (a && b) cout << "R";
```

```
else cout << "S";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 9

```
int a = -2, b = 0, c = 2, d = 1;
```

```
if (++a || ++b) cout << "Alpha";
```

```
else if (++c && ++d) cout << "Beta";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 10

```
int a = 10, b = 0, c = -10, d = 0;
```

```
if (b++ && a++) cout << "Up";
```

```
else if (++c || ++d) cout << "Down";
```

```
else cout << "Level";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 11

```
int a = 0, b = 0, c = 1, d = 1;
```

```
if (++a && ++b && --c) cout << "Left";
```

```
else if (d--) cout << "Right";
```

```
else cout << "Center";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 12

```
int a = 3, b = 3, c = 3, d = 3;
```

```
if (--a && --b && --c && --d) cout << "Go";
```

```
else cout << "Stop";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 13

```
int a = 1, b = 2, c = 3, d = 4;
```

```
if (a-- || b-- || c-- || d--) cout << "Stop";
```

```
else cout << "Go";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 14

```
int a = -1, b = -2, c = -3, d = 0;
```

```
if (++a || ++b || ++c) cout << "Negative";
```

```
else cout << "Zero";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 15

```
int a = 0, b = 1, c = 0, d = 1;
```

```
if (a++ && d++) cout << "Sun";
```

```
else if (b-- && c--) cout << "Moon";
```

```
else if (++a && ++d) cout << "Star";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 16

```
int a = 1, b = 1, c = 0, d = 0;
```

```
if (--a || b--) cout << "Code";
```

```
else if (c++ || d++) cout << "Debug";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 17

```
int a = 0, b = 0, c = 0, d = 0;
```

```
if (++a || ++b) cout << "First";
```

```
else if (++c || ++d) cout << "Second";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 18

```
int a = 5, b = 0, c = -5, d = 1;
```

```
if (b++ && a++) cout << "Cat";
```

```
else if (c++ || d++) cout << "Dog";
```

```
else cout << "Bird";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 19

```
int a = 1, b = 0, c = 1, d = 0;
```

```
if (a-- && b-- && c-- && d--) cout << "Win";
```

```
else if (++a && ++b) cout << "Lose";
```

```
else cout << "Draw";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Question 20

```
int a = 2, b = 2, c = -2, d = -2;
```

```
if (--a && --b && ++c && ++d) cout << "Magic";
```

```
else cout << "Muggle";
```

```
cout << "\n" << a << " " << b << " " << c << " " << d << endl;
```

Answer Key & Tracing Guide

#	Console Output	a b c d	Logic Summary
1	Banana	2 1 2 3	a=0 fails AND, a becomes 1. Next if : a=1 (True), a becomes 2. OR short-circuits.
2	Blue	0 0 0 0	Pre-decrements hit 0 (False). else if checks d (1=True), then d becomes 0.
3	Two	3 2 -2 0	b=0 fails AND, b becomes 1. Next if : ++b is 2 (True). ++a is 3 (True).
4	B	1 0 -1 -1	++a is 0, fails AND. Next if : a=0 fails, a→1. b=-1 is True, b→0. OR short-circuits.
5	No	1 1 1 1	All post-increments evaluate as 0 (False) before incrementing.

#	Console Output	a b c d	Logic Summary
6	All	4 4 4 4	All post-decrements evaluate as 5 (True) before decrementing.
7	Y	0 2 2 3	--a is 0, fails AND. Next if : c=3 (True), c→2. --d is 3 (True).
8	R	1 1 1 0	a=0 fails AND, a→1. c=0 fails OR, c→1. --d is 0, fails OR. a && b is True.
9	Alpha	-1 0 2 1	++a is -1 (True). OR short circuits. Remaining variables remain untouched.
10	Down	10 1 -9 0	b=0 fails AND, b→1. Next if : ++c is -9 (True). OR short-circuits.
11	Right	1 1 0 0	++a=1, ++b=1, but --c=0 fails the AND chain. Next if : d=1 (True), d→0.
12	Go	2 2 2 2	All pre-decrements result in 2 (True). Chain completes.
13	Stop	0 2 3 4	a=1 is True, a becomes 0. OR short-circuits immediately.
14	Negative	0 -1 -3 0	++a is 0 (False). ++b is -1 (True). OR short-circuits; c is untouched.

#	Console Output	a b c d	Logic Summary
15	Star	2 0 -1 2	a=0 fails AND, a→1. b=1(T), b→0. c=0(F), c→-1. Next if : ++a=2(T), ++d=2(T).
16	Code	0 0 0 0	--a is 0 (False). b-- is 1 (True), b becomes 0.
17	First	1 0 0 0	++a is 1 (True). OR short-circuits immediately.
18	Dog	5 1 -4 1	b=0 fails AND, b→1. Next if : c++ is -5 (True), c→-4. OR short-circuits.
19	Draw	1 0 1 0	a=1(T), a→0. b=0(F), b→-1. Next if : ++a=1(T), ++b=0(F). Both fail.
20	Magic	1 1 -1 -1	1, 1, -1, -1 are all non-zero (True). Complete chain executes.